



Fig. 2: Human brain

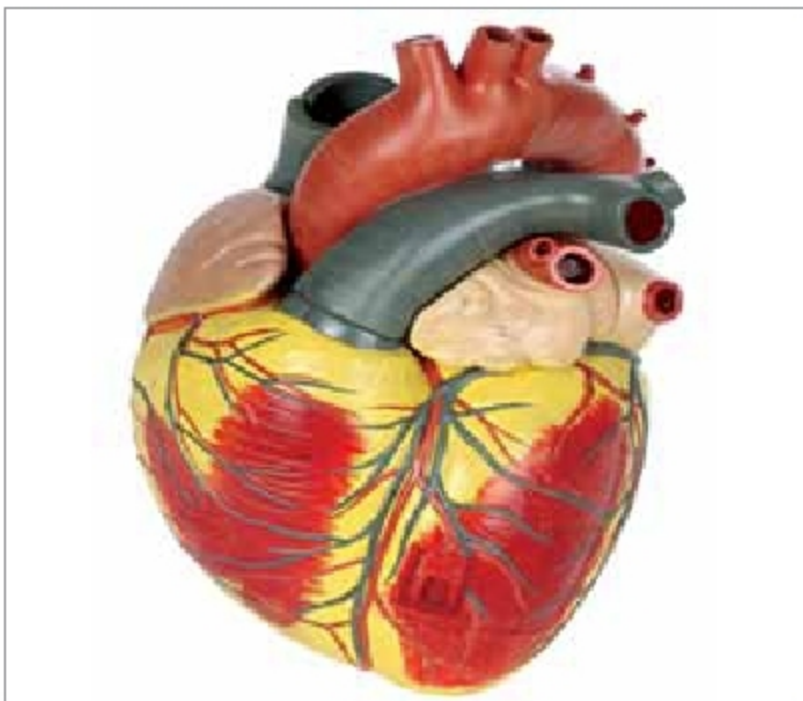


Fig. 3: Electronic heart



Fig. 4: Electronic skin

for long and healthy living. This article covers a brief history of developments related to different vital parts of a body like brain, skin, ear, eyes, nose, heart, liver, kidneys, hands, and legs.

However, electronic devices or implants may suffer electromagnetic interference, corrosion, and wear and tear with time. Also, due to repeated operations, these require batteries replacement. Therefore making corrosion-free and more durable electronic devices that can be powered using movement or other energy sources from the body could be the answer.

Human brain

Human brain is the most delicate and complicated organ that controls the functioning of the whole body. In case something goes wrong with the functioning of brain, doctors try to understand and repair it within some limits. Neural prostheses are a series of devices that can substitute motor, sensory, or cognitive modalities that might have been damaged as a result of an injury or disease.

Rather than replacing existing neural networks to restore function, these devices often serve by disrupting the output of existing malfunctioning nerve centres to eliminate symptoms. Scientists are working towards electronic brains. But brain is not just another organ as it contains all

your thoughts and memories. So, a new electronic brain in the head may not be that beneficial. However, this hasn't stopped scientists from growing actual human brains in a lab. Starting

with nothing but stem cells, scientists have managed to create brains equivalent to those of nine-week-old fetuses.

These miniature brains are the size of peas and are incapable of thought so far. The one thing keeping the brains from growing beyond this stage and becoming fully functional is that they have no blood supply. Though these brains aren't placed into anyone's body, these are proving a great resource for scientists investigating brain diseases.

Electronic heart

Implantable cardiac defibrillator (ICD) is an electronic device that is placed inside the body to track heart rhythm. It sends a low-intensity shock to the heart muscle if the rhythm becomes abnormal. Some ICDs use a pacing system instead of a shock.

An artificial heart can completely replace a diseased heart and get blood pressure back to normal. While it isn't meant to be a permanent replacement, it keeps people with heart failure alive until they receive an actual transplant. Cardiovascular-related artificial organs are implanted in cases where the heart, its valves, or another part of the circulatory system is not working properly.

Besides these, lab-grown and 3D-printed hearts are also being researched. Scientists have investigated stem cells' potential, and they have reached a major milestone after creating heart tissue that can beat on its own.

Researchers used stem cells made from the skin to make MCP, a special kind of cell that acts as a precursor to cardiovascular tissue. It was then placed on a 3D scaffold designed to support a mouse heart. Within twenty days, the new heart began beating at forty to fifty beats per minute. Though this heart is too weak to actually pump blood, the tissue has a lot of potential for repairing heart muscles that have suffered damage.

Electronic skin

Skin protects the whole body, making it the most easily damaged organ. When the skin is burnt or rips off, the main option right now is to graft some back on from elsewhere on the body. But an effective synthetic replacement of skin may not be that far. Skin-like

and enhance the understanding of living systems.

With all these developments, researchers are working on replacing every organ of the human body